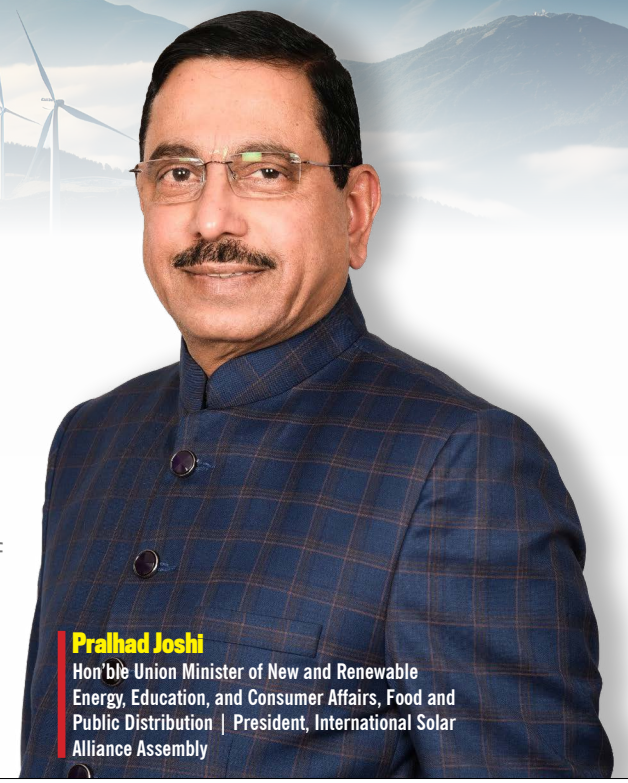
INDIA'S RENEWABLE ENERGY REVOLUTION

From surpassing global climate commitments ahead of schedule to building 500 GW of non-fossil capacity by 2030, India is accelerating its clean energy transition through solar power, indigenous manufacturing, grid modernisation and people-led initiatives.



Pralhad Joshi
Hon'ble Union Minister of New and Renewable Energy, Education, and Consumer Affairs, Food and Public Distribution | President, International Solar Alliance Assembly

1. India has crossed 300 GW and met its Paris commitments nine years early. What will define the next phase on the way to 500 GW by 2030?

Under the leadership of Hon'ble Prime Minister Shri Narendra Modi, India achieved its 40% non-fossil capacity commitment in 2021, nine years early, and its 50% commitment in June 2025, five years early.

Today, we have more than 308 GW of non-fossil capacity, including 168 GW of solar. Three priorities will define the next phase. The first is round-the-clock reliability through storage and transmission. The second is self-reliance across the manufacturing chain. The third is investment at scale, about USD 210 billion between FY2027 and FY2030. We will welcome global partners at the Bharat Renewable Energy Summit to be held from 2nd-5th November. When households and farmers produce their own power, the transition becomes a people's movement. That is how we will reach 500 GW by 2030 and build towards a Viksit Bharat.

2.As President of the ISA Assembly, what are your priorities for the 9th Session, and how can India help member countries move from solar ambition to bankable projects?

The Ninth Session of the ISA Assembly will focus on moving from ambition to action for the Global South. Priorities include mobilising finance at scale through instruments such as the Africa Solar Facility, building institutional capacity via the Global Capability Centre, ISA Academy and STAR-C network, advancing integrated solar ecosystems under Solarising Africa, and promoting innovation including AI for Energy. Through Country Partnership Frameworks, the International Solar Alliance is supporting the development of solar projects with a combined capacity of 3.16 GW across 20 countries, worth USD 842 million. India will share practical experience from PM Surya Ghar and PM-KUSUM so member countries can develop bankable projects that deliver energy access, livelihoods and jobs. One Sun must bring shared prosperity.

3.What policy and market measures will help storage and grid integration keep pace with generation, so clean power is available round the clock?

Solar power is available mainly during daylight hours, while electricity demand often rises in the evening. We must therefore expand storage and transmission alongside renewable generation. The Government has integrated storage into grid planning. Green Energy Corridor Phase-III, approved by the Union Cabinet on 30 September, provides for 50 GWh of battery energy storage capacity, with an outlay of ₹50,000 crore. It will strengthen intra-state transmission networks to evacuate up to 135 GW of renewable energy by 2032–33. With India's storage requirement projected at around 411 GWh by 2031–32, these measures will help us integrate more renewable energy while maintaining a reliable electricity supply. The Total project outlay for the scheme is Rs.1,86,405 crore.

4.How do you see India becoming a trusted, globally competitive supplier of clean energy equipment and green molecules?

India's manufacturing growth rests on consistent policy. Production Linked Incentives encouraged companies to build integrated factories in India. Approved lists of models and manufacturers ensured quality, and since June 2026 this requirement has extended to solar cells. This resulted in solar module capacity rising from about 8 GW before 2021 to 225 GW, and cell capacity from 3 GW to 37 GW. Module imports fell from USD 4.3 billion in 2023-24 to USD 0.9 billion in 2025-26. Similarly, in wind, India is the world's third-largest turbine maker and a net exporter of equipment. In batteries, 40 GWh of cell manufacturing capacity has been awarded. Under the National Green Hydrogen Mission, we aim to produce 5 million tonnes of green hydrogen a year by 2030. Consistent quality and stable policy will make India a trusted partner.

5.How is the government combining large-scale finance with distributed solutions like rooftop solar and agricultural solarisation, so the transition reaches every citizen?

Transparent bidding and stable policies have attracted about USD 164 billion into renewables since 2014. The government funds direct subsidies for citizens. This money lowers technology costs, allowing programs like PM Surya Ghar and PM-KUSUM to give households free electricity and turn farmers into clean energy producers. Essentially, big finance drives down prices, making rooftop and farm solar affordable for every doorstep.Alongside this, we have placed clean energy directly in citizens' hands. PM Surya Ghar: Muft Bijli Yojana now benefits more than 65 lakh households, and nearly 21 lakh families receive zero electricity bills. Under PM-KUSUM, around 30 lakh pumps have been solarised. Our Annadata is becoming an Urjadata. Large projects bring costs down, and distributed systems take those benefits to every doorstep.